Finalise ADVANCE TRAUMA SAP

SAP v0.7.0: comment review & decisions

2026-11-10

Context

- 89 comments on SAP v0.7.0 (2026-10-07)
- Today: make 11 substantive decisions (~5 min each)
- Rest is wording/clarification, handled as offline drafting

Trial status

- Batch 1 finished. First SDMC meeting before end of year
- Batch 2 ongoing
- Batch 3 starting next week (delayed)
- ~2500 patients included.

The 11 decisions

1. Interim analysis vs routine trial monitoring (p. 13)
2. Length-of-stay: counts, durations, or time-to-event? (pp. 16, 18, 21, 26)
3. Intercurrent-events table (pp. 18–21)
4. Missing data (pp. 27–28)
5. Adjust the primary model for randomisation factors (pp. 23, 27)
6. Small-sample inference & model-fit criteria (pp. 23, 27)
7. QoL & disability (p. 25)
8. Effect sizes (pp. 22–24)
9. Sensitivity analyses (pp. 28–30)
10. Subgroups (pp. 30–31)
11. Repeated ED presentations (p. 30)

1. Is "Interim analysis" really trial monitoring (§4.5, p. 13)

Decision: terminology, and what the half-way report contains.

- The document calls the half-way review an "interim analysis", but nothing in §4.5 is a stopping rule based on the treatment effect. The per-batch reports are really routine trial monitoring.

Suggestion:

- Call per batch summaries **"routine monitoring reports"**
- Call the half-way report **"interim analysis"** (included in protocol).
- Ask for a decision to continue or stop half-way.
- Exclude outcome comparisons and formal effect-based stopping rules.

2. Length-of-stay: counts, durations, or time-to-event? (§9.2.5–9.2.8, pp. 16, 18, 21, 26)

Decision: statistical model, effect measure, estimand wording (ED, hospital, ICU LOS).

- LOS is analysed with negative binomial regression, but the data are recorded as **date and time** (continuous durations).
- Options:
  - **gamma/log-link** reporting a ratio of expected LOS
  - **time-to-event** with death as a competing risk; either way the estimand (observed duration until exit) should be stated and

Suggestion:

- State the estimand as observed duration until exit an prefer the gamma/log-link model.
- Interpret alongside mortality and transfer frequencies.

3. Intercurrent-events table: what belongs there? (§8.1, pp. 18–21)

Decision: restructure Table 1 / simplify §8.1.

- The table currently mixes three different things: true intercurrent events, missing-data issues (loss to follow-up, outcomes that cannot be ascertained), and an outcome definition (ICU admission only if admission precedes death/transfer).

Suggestion:

- Remove loss to follow up from the list of intercurrent events
- Remove ICU outcome definition as well
- Add "treatment policy" wording (exposure already follows the scheduled transition date)

4. Missing data: replace the 10% rule? (§9.4, pp. 27–28)

Decision: primary estimator, role of multiple imputation, imputation specification.

- The 10%-missing threshold is arbitrary: bias depends on the missingness mechanism.

Suggestion:

- drop the 10% fixed-threshold framing
- pre-specify primary analysis as using all available observed outcome data
- use multiple imputation as a sensitivity analysis under a MAR assumption
- consider missingness separately for each outcome

5. Adjust the primary model for randomisation factors? (§9.1, pp. 23, 27)

Decision: include the cluster-size randomisation variable as a fixed covariate?

- Randomisation is covariate-constrained on anticipated cluster size, but the primary model does not adjust for it (it adjusts for time and cluster random effects only).

Suggestion: include anticipated cluster size as a fixed covariate (adjusting for randomisation factors is a reviewer expectation) and make the cluster + cluster-by-period random effects explicit in the text.

6. Small-sample inference & model-fit criteria (§9, pp. 23, 27)

Decision: pre-specified wording for corrections and acceptable fit.

- The SAP needs a pre-specified small-sample correction for the non-linear models (the choice affects SEs, CIs and p-values and must not be data-driven), plus criteria for acceptable fit beyond convergence.

Suggestion:

- Kenward–Roger for linear models and an objective fallback for GLMMs (the t(K−2) convention has worked well with few clusters)
- accept zero variance components (simpler structures as sensitivity)
- keep randomisation at hospital level, no unit-level random effect.

7. QoL & disability: separate or longitudinal? (§9.2.2–9.2.3, p. 25)

Decision: per-time-point analyses, or one joint longitudinal model?

- QoL and disability are measured at several post-discharge time points and the current text implies separate per-time-point analyses.

Suggestion: one **joint longitudinal model**, with time, time×treatment interaction and patient-level random effects, reporting the effect at each time point from the model.

8. Effect sizes: odds ratio, risk ratio, or marginal effect? (§8.4, §9.1, pp. 22–24)

Decision: the effect measure each outcome reports:

- We currently state that we will estimate both the OR and the ARD. Should we change this to reporting RR?
- The adherence proportion is reported as a logit-scale difference; an average marginal effect (the difference in expected percentage of checklist items completed) is easier to communicate.

Suggestion:

- Keep OR and ARD
- For adherence, report the **average marginal effect** as a difference in expected percentage points.
- Before finalising, clarify what "get Res on these" refers to (KH?)

9. Sensitivity analyses (§9.5, pp. 28–30)

Decision: the final list of sensitivity analyses, consistent with the protocol.

- The list is long and a prior paper was criticised for too many sensitivity analyses.

Currently specified analyses:

Sensitivity analysis	In the protocol?
9.5.1 Exposure by actual transition date	No (SAP addition)
9.5.2 Autoregressive within-cluster correlation	Yes
9.5.3 Random cluster $\times$ intervention	Yes
9.5.4 Time modelled with a spline	Yes
9.5.5 Lag & weaning effects	Yes
Multiple imputation under MAR	No (SAP missing-data plan)

Suggestion:

- keep the analyses to avoid a major protocol amendment.

10. Subgroups: age groups & geographical region (§9.7, pp. 30–31)

Decision: age categorisation; how the primary subgroup is handled.

- Four age groups risk some subgroups with very few events and the expected patient distribution is unknown.
- The final set of states (geographical region) is not fixed, yet region is currently the primary subgroup.

Suggestion:

- use two clinically justified age groups (cut-off fixed before analysis)
- for region, use zonal councils (Northern, Southern, Central, Eastern, Western, North Eastern)

11. Can a patient appear more than once? (§9, p. 30)

Decision: inclusion rule for repeated ED presentations; any within-patient correlation modelling.

- A patient can present to the ED more than once, but the models only include cluster-level random effects.

Suggestion: treat multiple presentations as independent because they are expected to be rare (alternative is to include only the first presentation per patient)